Python II module - Exercises

Pieter Kleer

Table of contents

Overview 2

Lecture 1: Sections 3.1-3.5 3

Lecture 2: Sections 3.6-3.7 and 4.1-4.3 6

Lecture 3: Section 5.1 11

Lecture 4: Section 5.2 23

Overview

This online document contains all the exercises corresponding to the lectures. Solutions are provided on a week-by-week basis in a different document.

Lecture 1: Sections 3.1-3.5

Make sure to import NumPy to be able to use all its functionality.

```
import numpy as np
```

Do not use for- or while-loops when answering the questions below.

Question 1

Create a 3×2 array M of ones, i.e., the matrix

```
[[1. 1.]
 [1. 1.]
 [1. 1.]]
```

Question 2

Create the array $x = [2, 4, 6, 8, ..., 100]$ once with `np.arange()`, and once with `np.linspace()`.

Question 3

Create the array $x = [-1, -0.9, \dots, -0.1, 0, 0.1, \dots, 0.9, 1]$.

Question 4

Consider the two-dimensional array M and answer the questions below. Recall that Python always starts counting at 0 when indexing.

```
M = np.array([[1,1,1,1,1],[2,1,2,1,2],[2,1,2,1,2],
              [2,1,2,1,2],[2,1,2,1,2],[1,1,1,1,1]])

print(M)
```

```
[[1 1 1 1 1]
 [2 1 2 1 2]
 [2 1 2 1 2]
 [2 1 2 1 2]
 [2 1 2 1 2]
 [1 1 1 1 1]]
```

a) Access the rows of M with odd indices $1, 3$ and 5. Your output should be

```
[[2 1 2 1 2]
 [2 1 2 1 2]
 [1 1 1 1 1]]
```

b) Index/access the rows and columns that form the submatrix consisting of elements equal to 2, i.e., obtain from M the matrix

```
[[2 2 2]
 [2 2 2]
 [2 2 2]
 [2 2 2]]
```

c) Access the submatrix consisting of the rows with indices $0, 1$ and 5 and columns with indices $1, 3, 4$, i.e.,

```
[[1 1 1]
 [1 1 2]
 [1 1 1]]
```

Question 5

Consider the following array.

```
x = np.array([3,6,4,5,5,5,1,4,2,9,6,7,11,10])

print(x)
```

```
[ 3  6  4  5  5  5  1  4  2  9  6  7 11 10]
```

a) Use a Boolean mask to access all elements whose value is less than or equal to 4.

b) Use a Boolean mask to access all elements whose value is in the interval $[5, 10]$.

c) Use a Boolean mask to access all elements whose value is in the interval $[2, 4]$ or $[6, 9]$.

Question 6

Take the array $x = [0, 1, 2, 3]$ and convert it to

$$\begin{bmatrix} 0 & 1 & 2 & 3 & 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 3 & 0 & 1 & 2 & 3 \end{bmatrix}.$$

Question 7

Construct the following matrix M using appropriate NumPy functions starting from the array $[1, 2, 4]$:

$$\begin{bmatrix} 1 & 1 & 2 & 2 & 4 & 4 \\ 1 & 1 & 2 & 2 & 4 & 4 \end{bmatrix}.$$

Question 8

Write a function `blocks(m,n)` that, for given integer inputs m and n, returns an $(m+n) \times (m+n)$ matrix that contains an $m \times m$ block of ones on the top left, and an $n \times n$ block of ones on the bottom right (and zeros elsewhere). Use `hstack()` and `vstack()` in your solution.

For $m = 2$ and $n = 3$, this results in the matrix

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

Question 9

Write a function `checkerboard(n)` that returns for an integer n a checkerboard pattern of zeros and ones of size $n \times n$ (see examples below; the top-left element is always a 1).

For $n = 5$ and $n = 6$, the matrices should look as follows:

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{bmatrix}.$$

Question 10

Compute the anti-diagonal of the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 5 & 6 & -7 \\ -9 & 10 & 11 \end{bmatrix}$$

using `np.rot90()`. The anti-diagonal is obtained by going from the bottom-left to the top-right element, i.e., $[-9, 6, 3]$ in this case (and not $[3, 6, -9]$). Have a look at the documentation of `np.rot90()` for details of how this function works.

Lecture 2: Sections 3.6-3.7 and 4.1-4.3

Make sure to import NumPy to be able to use all its functionality. We also add a command that displays floating-point numbers in arrays to three decimal places. You do not have to know this command.

```
import numpy as np
np.set_printoptions(precision=3, floatmode="fixed")
```

Do not use for- or while-loops when answering the questions below.

Question 1

Create the two-dimensional array

$$M = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \\ 17 & 18 & 19 & 20 \\ 21 & 22 & 23 & 24 \end{bmatrix}$$

by combining two functions seen in Chapter 3.

Question 2

Create the array

$$x = [1, 5, 9, 2, 6, 10, 3, 7, 11, 4, 8, 12]$$

from the first three rows of M (using reshaping functionality in combination with transposing a matrix).

Question 3

Implement the following function

$$f(x) = \begin{cases} -x + 3 & \text{if } x < 0 \\ x^2 + 3 & \text{if } 0 \leq x < 1 \\ \sqrt{x^2 + 3} + 2 & \text{if } x \geq 1 \end{cases}$$

so that it can handle both single numbers x and one-dimensional arrays $x \in \mathbb{R}^n$ as input. You might want to look at the Heaviside function example in Section 4.1 for some inspiration.

Your function should give the following output on the given input x.

```
x = np.arange(-5,6)
print(f(x))
```

```
[8.000 7.000 6.000 5.000 4.000 3.000 4.000 4.646 5.464 6.359 7.292]
```

Question 4

We will write a function that can compute the cumulative mean of a one-dimensional array. Take $n = 10$ (define this as a variable in your script).

a) Create the array $y = [1, 1/2, 1/3, \dots, 1/(n-1), 1/n]$ using `np.arange()` in combination with a division.

b) By combining your solution in part a) with `np.cumsum()`, create a function `cum_mean()` that takes as input a one-dimensional array $x = [x_0, \dots, x_{n-1}]$ and outputs the cumulative means of the array. This is the array that has at position i the value

$$\frac{1}{i+1} \sum_{j=0}^{i} x_j$$

for $i = 0, \dots, n-1$.

It should give the following output on the given input x.

```
# Some test data
x = np.array([1,4,2,5])
print(cum_mean(x))
```

```
[1.000 2.500 2.333 3.000]
```

c) Vectorize your function from part b) so that it takes as input two-dimensional arrays, and outputs the cumulative mean of every row of the two-dimensional array. Your function should give the following output on the given input matrix M.

```
# Some test data
M = np.array([[1,4,2,5],[1,10,12,8],[-1,9,3,-10]])
print(cum_mean(M))
```

```
[[ 1.000  2.500  2.333  3.000]
 [ 1.000  5.500  7.667  7.750]
 [-1.000  4.000  3.667  0.250]]
```

Question 5

Consider the following function

$$g(x_0, \dots, x_{n-1}) = \sum_{i=0}^{n-1} \sin(x_i) \cdot (x_i)^{2 \cdot i}$$

that takes as input an array $x = [x_0, \dots, x_{n-1}] \in \mathbb{R}^n$ and outputs $g(x)$.

a) Implement the function g. It should give the following output on the given input x.

```
# Some input data
x = np.array([1,4,2,6,4,5])
print(g(x))
```

```
-9427125.80618379
```

b) Vectorize the function g so that it can take as input a two-dimensional array, and return the function value $g(x)$ for every row x of the array.

It should give the following output on the given input M.

```
# Some input data
M = np.array([[1,4,2,6,4,5],[1,4,2,6,4,5],[7,4,9,6,3,5]])
print(g(M))
```

```
[-9427125.806 -9427125.806 -9373912.933]
```

Question 6

Write a function `geom(x)` that takes as input a two-dimensional array, and outputs the geometric mean of every column of the array. For an array $x = [x_0, \dots, x_{n-1}] \in \mathbb{R}^n_{\geq 0}$, the geometric mean is defined as

$$\left(\prod_{i=0}^{n-1} x_i \right)^{1/n}$$

It should give the following output on the given input M.

```
# Some input data
M = np.array([[1,4,2,6,4,5],[1,4,3,7,1,5],[1,4,2,6,8,50]])
print(geom(M))
```

```
[ 1.000  4.000  2.289  6.316  3.175 10.772]
```

Question 7

In this exercise we will normalize the data in an array, so that all entries are between 0 and 1.

a) Write a function `normalize()` that normalizes an array $x = [x_0, \dots, x_{n-1}]$ by replacing every entry x_i by

$$\frac{x_i - x_{\text{min}}}{x_{\text{max}} - x_{\text{min}}}$$

where $x_{\text{min}} = \min_i x_i$ and $x_{\text{max}} = \max_i x_i$. You may assume that $x_{\text{max}} \neq x_{\text{min}}$ to avoid dividing by zero. *You could add a check for this case into your function, but is not needed.*

It should give the following output on the given input x.

```
# Some input data
x = np.array([1,4,2,-6,4,5])
print(normalize(x))
```

```
[0.636 0.909 0.727 0.000 0.909 1.000]
```

b) Vectorize your function so that it can normalize every column of a two-dimensional array using the formula in part a). You may, again, assume that the maximum and minimum element in each column are different.

It should give the following output on the given input M.

```
# Some input data
M = np.array([[1,4,2,-6,4,5],[-4,3,5,1,3,2],[9,8,7,6,5,4]])
print(normalize(M))
```

```
[[0.385 0.200 0.000 0.000 0.500 1.000]
 [0.000 0.000 0.600 0.583 0.000 0.000]
 [1.000 1.000 1.000 1.000 1.000 0.667]]
```

Question 8

In this exercise we will implement a different type of data normalization.

a) Write a function `normal()` that normalizes a two-dimensional array (matrix) M such that the entries in each row have mean 0 and standard deviation 1. You can do this by subtracting each row's mean from every element in that row, and dividing every element by the standard deviation of the row. You may assume that the standard deviation of a row is not equal to zero.
Hint: A good understanding of the notion of broadcasting, see, e.g., Section 4.1, might be useful here.

It should give the following output on the given input M.

```
# Some test data
M = np.array([[ 1,  2,  3,  0],
       [ 5,  6,  -7,  0],
       [ -9, 10, 11, 0],
       [13, -13, 15, 0],
       [17, 18, 19, 0],
       [-21, -22, -23, 0]])
print(normal(M))
```

```
[[-0.447  0.447  1.342 -1.342]
 [ 0.777  0.971 -1.554 -0.194]
 [-1.472  0.858  0.981 -0.368]
 [ 0.822 -1.488  1.000 -0.333]
 [ 0.447  0.575  0.703 -1.725]
 [-0.471 -0.576 -0.680  1.727]]
```

b) Verify that the rows have mean 0 using the `np.mean()` function from NumPy.

c) Verify that the rows have standard deviation 1 using the `np.std()` function from NumPy.

Lecture 3: Section 5.1

Make sure to import NumPy to be able to do all the exercises.

```
import numpy as np
np.set_printoptions(precision=3, floatmode="fixed", suppress=True)
```

Question 1

Suppose that $A \in \mathbb{R}^{m \times n}$ is matrix and $b \in \mathbb{R}^m$ a vector. If $m \geq n$ and the columns of A are linearly independent, i.e., if the column rank of A is n, then the unique least squares solution to

$$\min_x ||Ax - b||_2.$$

is given by $x^* = (A^T A)^{-1} A^T b$. (The combination of $m \geq n$ and A having linearly independent columns guarantees that $(A^T A)^{-1}$ exists.)

Write a function that takes as input a matrix A and vector b. It should return the least squares solution x^* if A has full rank, and a message saying that A does not have full column rank if not. You may use an if/else-statement for this case distinction.

It should give the following output on the inputs below.

```
A = np.array([
 [0.61776137, 0.9880466,  0.93526716],
 [0.25839379, 0.66983101, 0.55140378],
 [0.42598981, 0.47034738, 0.67713771],
 [0.05808045, 0.60282648, 0.968094  ],
 [0.6152295,  0.18224588, 0.23185116]])

b = np.array([
 [0.68873052],
 [0.01618962],
 [0.67498566],
 [0.73718501],
 [0.0448842 ]])

print(least_squares(A,b))
```

```
[[ 0.052]
 [-0.769]
 [ 1.329]]
```

```
A = np.array([
 [0.61776137, 0.9880466,  0.9880466],
 [0.25839379, 0.66983101, 0.66983101],
 [0.42598981, 0.47034738, 0.47034738],
 [0.05808045, 0.60282648, 0.60282648],
 [0.6152295,  0.60282648, 0.60282648]])

b = np.array([0.68873052,0.01618962, 0.67498566, 0.73718501, 0.0448842])

# Columns are not linearly independent
print(least_squares(A,b))
```

```
Matrix A does not have full column rank
```

Question 2

In this exercise we are going to implement Cramer's rule for solving the system $Ax = b$ when $A \in \mathbb{R}^{n \times n}$ is an invertible matrix. Cramer's rule determines the solution $x = [x_0, \dots, x_{n-1}]$ by

$$x_i = \frac{\det(A_i)}{\det(A)} \quad \text{for } i = 0, \dots, n-1,$$

where A_i is the matrix A in which column i is replaced by b.

Write a function `cramer(A,b)` that computes the solution x according to Cramer's rule (you may use one for-loop in your implementation).

Your function should give the following output on the given input.

```
A = np.array([
 [0.61776137, 0.9880466,  0.93526716],
 [0.25839379, 0.66983101, 0.55140378],
 [0.42598981, 0.47034738, 0.67713771]
 ])

b = np.array([0.68873052, 0.01618962, 0.67498566])

# Solution with Cramer's rule
print(cramer(A,b))

# Solution with np.linalg.solve() for comparison
print(np.linalg.solve(A,b))
```

```
[ 2.884 -0.970 -0.144]
[ 2.884 -0.970 -0.144]
```

Question 3

Assume we are given a matrix $A \in \mathbb{R}^{m \times n}$ with the property that every subset of m columns of A is linearly independent. This is, e.g., typically the case if the entries in A were randomly generated (we will see this later in the course).

One way to obtain a solution to an underdetermined system ($n > m$) $Ax = b$ is to pick a set of indices $R \subseteq \{0, \dots, n-1\}$ of size $|R| = m$. Set $x_i = 0$ for all $i \notin R$ (there are $n - m$ such variables), and determine the values in $x_R = (x_i)_{i \in R}$ by solving the system $Bx_R = b$, where B is the matrix containing the columns of A corresponding to the indices in R.

Write a function `underdetermined(A,b,R)` that takes as input arrays A, b and a set R (using a list) as above, and returns a solution $x \in \mathbb{R}^n$ to $Ax = b$ satisfying the above description.

Your function should give the following output on the given input.

```
A = np.array([
 [0.83, 0.43, 0.16, 0.01, 0.86, 0.87, 0.36, 0.68],
 [0.29, 0.81, 0.78, 0.97, 0.2,  0.28, 0.44, 0.07],
 [0.18, 0.81, 0.91, 0.88, 0.54, 0.89, 0.53, 0.88]])

b = np.array([0.68873052, 0.01618962, 0.67498566])

R = [0,4,7]

# Determine solution
x = underdetermined(A,b,R)
print(x)

# Verify correctness
print(A @ x - b) # All numbers almost zero (e-16 is numerical inaccuracies)
```

```
[-3.163  0.000  0.000  0.000  5.314  0.000  0.000 -1.847]
[0.000 0.000 0.000]
```

Question 4

In this exercise, we will look into the basics of principal component analysis (PCA); we emphasize that this is not a complete explanation, but we hope it helps.

We are given an $m \times n$ matrix A containing $n \geq 2$ data points that each have m features. That is, every column of A corresponds to a data point.

We will be working with the dataset below, in which we have $n = 30$ data points, each with $m = 3$ features: weight, height and IQ. In practice, datasets are typically much larger, but we keep the set small here to be able to visualize it.

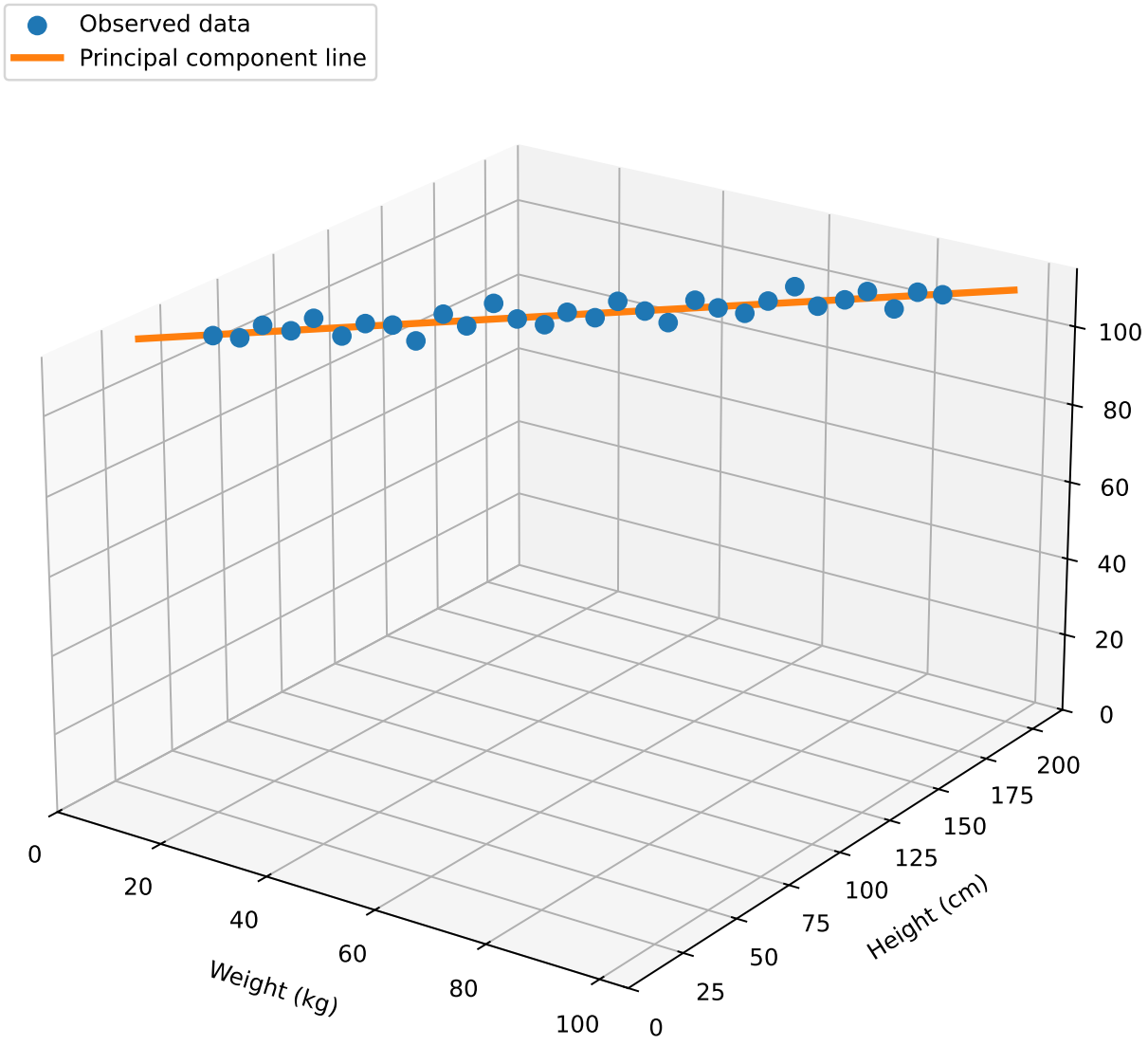
```
# Rows: weight (kg), height (cm), IQ; columns: 30 data points
A = np.array([
    [10.0, 12.8, 15.5, 18.3, 21.0, 23.8, 26.6, 29.3, 32.1, 34.8,
```

```
       37.6, 40.3, 43.1, 45.9, 48.6, 51.4, 54.1, 56.9, 59.7, 62.4,
       65.2, 67.9, 70.7, 73.4, 76.2, 79.0, 81.7, 84.5, 87.2, 90.0],

      [50.0, 55.2, 59.1, 65.0, 68.8, 74.6, 78.5, 84.3, 88.0, 94.0,
       97.8, 103.6, 107.5, 113.2, 117.0, 123.0, 126.8, 132.5, 136.4, 142.2,
       146.0, 151.9, 155.7, 161.5, 165.4, 171.2, 175.0, 180.9, 185.1, 190.0],

      [110.0, 109.0, 112.0, 110.0, 113.0, 108.0, 111.0, 110.0, 106.0, 112.0,
       109.0, 114.0, 110.0, 108.0, 111.0, 109.0, 113.0, 110.0, 107.0, 112.0,
       110.0, 108.0, 111.0, 114.0, 109.0, 110.0, 112.0, 107.0, 111.0, 110.0]
  ])
```

Although every data point is three-dimensional, it turns out these points, roughly speaking, can be "summarized" in a lower dimension. Indeed, the line in the figure below describes the data quite well and is one-dimensional.



The rough idea is to establish a description of the so-called principal component line, which mathematically speaking is a one-dimensional affine subspace

$$\begin{bmatrix} \bar{\mu}_1 \\ \bar{\mu}_2 \\ \bar{\mu}_3 \end{bmatrix} + \text{span}\{v_0\}$$

where the $\bar{\mu}_i$'s will be the row averages of A and $v_0 \in \mathbb{R}^3$ is a vector we determine later. Every data point column A_i of A can be approximated on this line by

$$\hat{A}_i = \begin{bmatrix} \bar{\mu}_0 \\ \bar{\mu}_1 \\ \bar{\mu}_2 \end{bmatrix} + v_0 \cdot Z_i$$

for some $Z_i \in \mathbb{R}$. This Z_i is the one-dimensional representation of $\hat{A}_i$, which in turn is close to A_i.

More generally, it turns out that there are many datasets for which the m features can be summarized in a k-dimensional subspace span$\{v_0, \dots, v_{k-1}\} \subset \mathbb{R}^m$ with $k \ll m$. Every data point is approximated by an affine linear combination of k lines, i.e.,

$$\hat{A}_i = \begin{bmatrix} \bar{\mu}_0 \\ \vdots \\ \bar{\mu}_{m-1} \end{bmatrix} + V_k Z_i = \begin{bmatrix} \bar{\mu}_0 \\ \vdots \\ \bar{\mu}_{m-1} \end{bmatrix} + \sum_{j=0}^{k-1} z_{ij} v_j$$

with $V_k = [v_0 \ \dots \ v_{k-1}]$ having the v_i's as columns, and $Z_i = [z_{i0} \dots z_{i(k-1)}]^T \in \mathbb{R}^k$ the k-dimensional column vector representation of $\hat{A}_i$.

This type of dimensionality reduction in data is often useful for advanced machine learning algorithms so that they can run more efficiently by using the lower-dimensional Z_i points instead of the original higher-dimensional A_i data points.

In the procedure that follows, we will show how this lower-dimensional subspace can be constructed, using eigenvalues and eigenvectors. For a given feature, the sample mean $\bar{\mu}_i$ of row i in A is defined as the average of all the values in that row. By subtracting from all elements in a row the row mean, we obtain the matrix

$$B = A - \bar{\mu}\mathbf{1}^T,$$

where $\mathbf{1}^T \in \mathbb{R}^n$ is an all-ones row vector (so that $\bar{\mu}\mathbf{1}^T$ is indeed a matrix). Next, the (symmetric) $m \times m$ covariance matrix S is defined by

$$S = \frac{1}{n-1} BB^T$$

a) Write a function `covariance()` that takes as input the matrix A and outputs the matrix S.

Testing your function on the data points in the array A given above should yield the following results.

```
# Covariance matrix
S = covariance(A)
print('S = \n',S, '\n')
```

```
S =
 [[ 589.528 1032.059   -0.421]
 [1032.059 1807.019   -0.712]
 [  -0.421   -0.712    4.097]]
```

The covariance matrix contains for $i = 0, \dots, m-1$ on its main diagonal the sample variances

$$S_{ii} = \frac{1}{n-1} \sum_{j=0}^{n-1} (A_{ij} - \bar{\mu}_i)^2$$

and on its off-diagonal elements the covariance of features i and j at position (i, j). The trace of S, the total variance in the data, is equal to the sum of the eigenvalues of the matrix S, because of the spectral decomposition (recall from your Linear Algebra course)

$$S = \sum_{i=0}^{m-1} \lambda_i v_i v_i^T$$

where λ_i are the eigenvalues of S and v_i the eigenvector corresponding to λ_i for $i = 0, \dots, m-1$, with the v_i forming an orthonormal basis.

b) Write a function `ordered_components()` that takes as input a symmetric matrix S and outputs a vector containing the eigenvalues in descending order (largest to smallest), and a matrix containing the corresponding eigenvectors ordered according to the order of the eigenvalues.

As an example, if we have three eigenvalues $\{\lambda_0, \lambda_1, \lambda_2\}$ with $\lambda_2 \geq \lambda_0 \geq \lambda_1$, then the output should be $[\lambda_2, \lambda_0, \lambda_1]$ and the matrix should be $[v_2\ v_0\ v_1]$ where the v_i are the eigenvectors corresponding to eigenvalues λ_i.
Hint: The function `np.argsort()` *from Chapter 4 might be useful here.*

Testing your function on the computed S in part a) should give the following results.

```
# Ordered eigenvalues and eigenvectors
S_values, S_vectors = ordered_components(S)
print('Eigenvalues are \n', S_values, '\n')
print('Eigenvectors are \n', S_vectors, '\n')
```

```
Eigenvalues are
 [2396.487    4.096    0.059]

Eigenvectors are
 [[ 0.496 -0.002  0.868]
 [ 0.868  0.002 -0.496]
 [-0.000  1.000  0.003]]
```

The covariance matrix is a sum of rank-one matrices $\lambda_i v_i v_i^T$. Each eigenvector gives a principal direction, and its eigenvalue gives the variance along that direction. We will next try to understand which eigenvectors contributes most to the data, by looking at the impact of their eigenvalues.

c) Write a function `contributions()` that takes a one-dimensional array $x = [x_0, \ldots, x_{n-1}]$ as input and outputs the numbers

$$y_i = \frac{\sum_{j=0}^{i} x_j}{\sum_{j=0}^{n-1} x_j}$$

for $i = 0, \ldots, n-1$, i.e., y_i is the fraction that the first $i+1$ numbers contribute to the sum of all numbers in x.

Applying the function to the output of part b) should give the following.

```
# Contributions of the k largest eigenvalues
k_largest = contributions(S_values)
print('Contribution of the k largest eigenvalues \n', k_largest)
```

```
Contribution of the k largest eigenvalues
 [0.998 1.000 1.000]
```

Further explanation

From the outputs above, we can see that the largest eigenvalue λ_0 is much larger than the others, and accounts for almost all of the total variance (sum of the eigenvalues), i.e., $\lambda_0/(\lambda_0+\lambda_1+\lambda_2) \approx 1$. This means that

$$S = \lambda_0 v_0 v_0^T + \lambda_1 v_1 v_1^T + \lambda_2 v_2 v_2^T \approx \lambda_0 v_0 v_0^T,$$

and the data points A_i in A are approximately contained in the one-dimensional (affine) subspace $\bar{\mu} + \text{span}\{v_0\} \subset \mathbb{R}^3$ with v_0 the eigenvector corresponding to λ_0. The approximation of A_i is given by

$$\hat{A}_i = \bar{\mu} + v_0 Z_i$$

where Z_i is the i-th entry in the row vector $Z = v_0^T B \in \mathbb{R}^n$, with B the centered data matrix.
More generally, if the eigenvalues of S are $\lambda_0 \geq \lambda_1 \geq \ldots \lambda_{m-1}$, and k is the smallest number of component so that the k largest eigenvalues account for almost all of the total variance, i.e.,

$$\frac{\lambda_0 + \cdots + \lambda_{k-1}}{\lambda_0 + \lambda_1 + \cdots + \lambda_{m-1}} \approx 1$$

then the data points are approximately contained in the affine subspace $\bar{\mu} + \text{span}\{v_0, \ldots, v_{k-1}\}$ as the columns of

$$\hat{A} = \bar{\mu}\mathbf{1}^T + V_k Z$$

where the columns of $Z = V_k^T B \in \mathbb{R}^{k \times n}$, with V_k the matrix containing as columns the eigenvectors $v_0, \ldots, v_{k-1}$, are the k-dimensional representations of the m-dimensional columns of $\hat{A}$.

Question 5

Following the final explanation in Question 4, we can ask ourselves the following question: If we start out with a data array containing m features, what is the smallest $1 \leq k \leq m$ that represents the data well enough? Mathematically speaking, we want the chosen k largest eigenvalues, and their eigenvectors that will form the k-dimensional (affine) subspace that the data is summarized in, to account for "almost all" variance (the sum of the eigenvalues) of the problem.

To formalize what we mean by "almost all" variance of the data, we can introduce a parameter $\delta \in [0, 1]$ that models what fraction of the total variance we want to recover; this value should be somewhat close to 1. For example if $\delta = 0.95$, this means we want to recover 95% of the total variance.

To summarize, we choose k as the smallest ℓ for which the ratio above is at least δ, i.e.,

$$k(\delta) = \min\left\{\ell \in \{1, \ldots, m\} : \frac{\lambda_0 + \cdots + \lambda_{\ell-1}}{\lambda_0 + \lambda_1 + \cdots + \lambda_{m-1}} \geq \delta\right\}.$$

a) Write a function `dim_reduction()` that takes as input a data array $A \in \mathbb{R}^{m \times n}$ and parameter $\delta \in [0, 1]$. It should output the lower-dimensional data set $Z = V_k^T B$ with V_k and B as described in Question 4.
To summarize, the function should repeat the steps of Question 4, for which you use calls to the functions created there, determine $k(\delta)$ and then compute Z.

b) Apply your function to the data array A below with $\delta = 0.95$.

```
A = np.array([
    [34.25, 38.20, 41.68, 39.56, 43.65, 45.79,
     39.20, 40.19, 42.25, 44.66, 44.95, 48.48,
     36.54, 40.47, 41.68, 45.07, 48.68, 51.21,
     41.52, 43.48, 45.92, 46.77, 49.67, 51.33,
     43.33, 48.19, 50.80, 49.55, 50.30, 55.34],

    [0.00, 22.86, 43.93, 50.56, 70.01, 90.21,
     10.01, 28.56, 42.60, 61.24, 73.97, 89.12,
     7.38, 26.77, 40.56, 49.11, 78.29, 95.62,
     1.18, 26.94, 48.22, 63.22, 77.63, 92.08,
     7.64, 16.57, 45.19, 62.76, 77.39, 99.98],

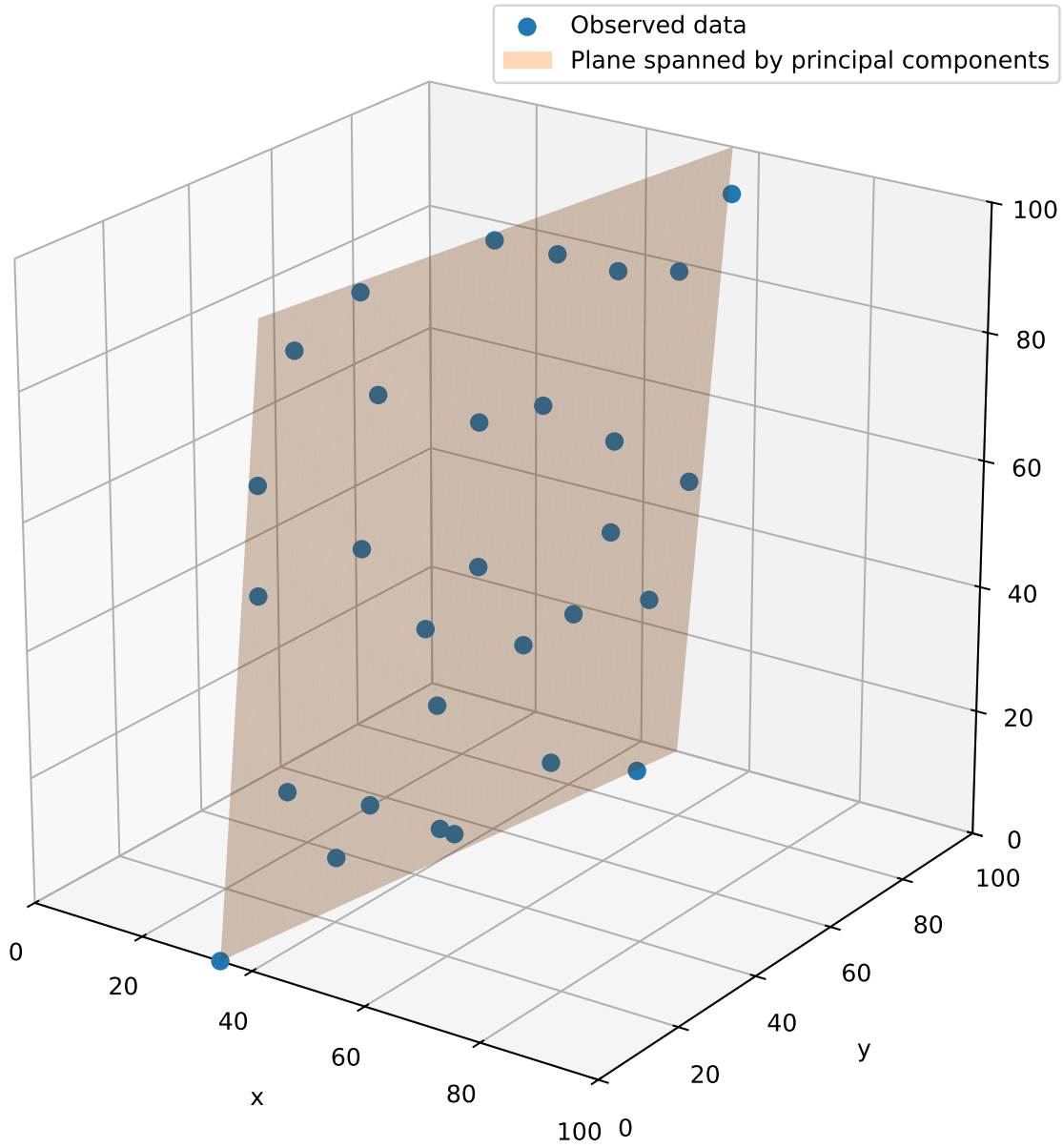
    [0.00, 8.72, 6.40, 2.52, 8.15, 0.00,
     24.20, 15.53, 26.90, 30.68, 31.40, 29.44,
     54.74, 56.46, 39.63, 47.34, 44.05, 47.20,
     74.62, 80.60, 70.36, 68.51, 59.03, 81.89,
     93.03, 100.00, 100.00, 92.63, 85.98, 92.63]
])

delta = 0.95

print(dim_reduction(A,delta))
```

```
[[ 41.512  36.028  41.200  45.665  42.831  53.546  19.237  30.164  20.818
   19.569  20.440  24.536 -11.510 -10.499   7.933   1.648   8.783   8.007
  -31.495 -34.087 -21.148 -17.410  -6.024 -26.715 -48.751 -54.131 -50.404
  -41.075 -32.645 -36.021]
 [ 50.874  28.781   9.414   2.896 -15.555 -33.806  39.907  23.041   8.788
   -9.080 -21.409 -35.069  38.795  20.755   9.362   1.095 -25.996 -42.506
   43.959  18.875  -0.144 -14.286 -26.624 -42.512  36.284  28.061   0.954
  -15.608 -28.935 -50.309]]
```

As can be seen from the output, the data is reduced from points in $\mathbb{R}^3$ to points in $\mathbb{R}^2$, so a dimensionality reduction from $m = 3$ to $k = 2$. The figure below also illustrates this. Furthermore, I hope you agree that there is no way to nicely capture the data in a one-dimensional line, i.e., reduction with $k = 1$.



Question 6

A more general version of the spectral decomposition of a symmetric $n \times n$ matrix is given by the singular value decomposition (SVD) of a general, not necessarily square matrix $A \in \mathbb{R}^{m \times n}$. Assuming $m \leq n$,

$$A = U\Sigma V^T = \underbrace{\begin{bmatrix} u_0 & \cdots & u_{m-1} \end{bmatrix}}_{m \times m} \underbrace{\begin{bmatrix} \sigma_0 & & 0 \\ & \ddots & \\ 0 & & \sigma_{m-1} \end{bmatrix}}_{m \times m} \underbrace{\begin{bmatrix} v_0^T \\ \vdots \\ v_{m-1}^T \end{bmatrix}}_{m \times n} = \sum_{i=0}^{m-1} \sigma_i u_i v_i^T .$$

with $\sigma_0 \geq \sigma_1 \geq \cdots \geq \sigma_{(m-1)} \geq 0$. For example,

$$A = \begin{bmatrix} 22 & 13 & 2 \\ 4 & 16 & 14 \end{bmatrix} = \underbrace{\begin{bmatrix} 0.8 & -0.6 \\ 0.6 & 0.8 \end{bmatrix}}_{U\colon 2\times 2} \underbrace{\begin{bmatrix} 30 & 0 \\ 0 & 15 \end{bmatrix}}_{\Sigma\colon 2\times 2} \underbrace{\begin{bmatrix} \frac{2}{3} & \frac{2}{3} & \frac{1}{3} \\ -\frac{2}{3} & \frac{1}{3} & \frac{2}{3} \end{bmatrix}}_{V^T\colon 2\times 3}.$$

The arrays U, Σ and V^T can be computed using the `np.linalg.svd()` function. Check out the documentation to see how.

Singular value decomposition can play an important role in image compression techniques, where the goal is to create an image with much smaller file size at the cost of some image detail.

Each entry of A gives the grayscale intensity of a pixel: 0 is black and 255 is white. A compressed image can be obtained by replacing A by the lower-rank approximation

$$A_k = U_k \Sigma_k V_k^T = \underbrace{\begin{bmatrix} u_0 & \cdots & u_{k-1} \end{bmatrix}}_{m\times k} \underbrace{\begin{bmatrix} \sigma_0 & & 0 \\ & \ddots & \\ 0 & & \sigma_{k-1} \end{bmatrix}}_{k\times k} \underbrace{\begin{bmatrix} v_0^T \\ \vdots \\ v_{k-1}^T \end{bmatrix}}_{k\times n} = \sum_{i=0}^{k-1} \sigma_i u_i v_i^T.$$

a) Write a function `compression()` that takes as input a two-dimensional array A containing pixel values, and a one-dimensional array containing integer values for k. It should return a list whose entries correspond to the arrays A_k as described above. You can use a for-loop to iterate over the values in the one-dimensional array.

b) Test your function by doing the following:

- Store the file grayscale_image.png in the folder with the `.py` file in which you are making these questions.
- Copy the test code below into your `.py` file, place your `compression()` function at the indicated location, and run your code. *You do not have to understand the additional image-handling code lines.*

  If all went well, multiple `.png` files should have been created, in the folder containing your files for this question, with names `image_k_xxx.png` where `xxx` represents the value k from the array `k_values` that has been used to create the image.

  Compare the file sizes with one another and with the original image's file size. What do you conclude?

```
# Packages for importing and working with images
from pathlib import Path
from PIL import Image

import numpy as np

# 1. Import the image
image_path = Path("grayscale_image.png")
```

```
with Image.open(image_path) as image:
    # Convert image to array with grayscale ("L") values between 0 and 255
    A = np.asarray(image.convert("L"), dtype=float)

# 2. Place function compression() here


# 3. Set k_values and determine A_k matrices
k_values = np.array([1, 5, 10, 20, 50, 100])
Ak_matrices = compression(A, k_values)


# 4. Save each reduced image beside the original image
for i in range(np.size(k_values)):
    # Round values in A_k to closest value in {0,1,2,3,...,255}
    pixels = np.rint(np.clip(Ak_matrices[i], 0, 255)).astype(np.uint8)

    # Create file name to store reduced image in
    png_path = image_path.parent / f"image_k_{k_values[i]:03d}.png"

    # Convert pixel values to .png image
    Image.fromarray(pixels).save(png_path)
```

As a final remark, we note that there is not a rigorous guarantee that the file size of a compressed image will always go down as k becomes smaller, because this process also has to do with how the matrix A_k is converted to a `.png` file. However, these are technicalities beyond the point of this exercise.

Lecture 4: Section 5.2

Make sure to import NumPy and the relevant functions from Gurobi.

```
import numpy as np
from gurobipy import Model, GRB, quicksum
```

Question 1

Médecins sans Frontières (MSF) (Dutch: Artsen zonder Grenzen) wants to build medical kits for use in region experiencing an epidemic. The following constraints should be taken into account.

- MSF has raised € 6400 to build two types of medical kits: vaccination and surgical kits. Assembling a surgical kit costs € 320; a vaccination kit € 210.
- MSF has a total of 80 labor hours available during which employees can build these kits. Building a surgical kit requires 2.5 labor hours; a vaccination kit requires 4.5 labor hours.
- At most 15 vaccination kits can be built.

A linear optimization problem that maximizes the total number of produced kits is given below. The (possibly fractional) decision variables are the number of surgical kits (x_1) and the number of vaccination kits (x_2) to assemble.

$$
\begin{array}{llrcrcrl}
\text{max} & z = & x_1 & + & x_2 & & & \textit{total number of kits} \\
\text{s.t.} & & 320x_1 & + & 210x_2 & \leq & 6400 & \textit{budget constraint} \\
& & 2.5x_1 & + & 4.5x_2 & \leq & 80 & \textit{labor hours constraint} \\
& & & & x_2 & \leq & 15 & \textit{at most } 15 \textit{ vac. kits} \\
& & x_1, & & x_2 & \geq & 0 & \textit{nonneg. constraints}
\end{array}
$$

a) Implement this model in Python with a Model object called `msf`, with (continuous) decision variables whose `name` keyword arguments are `'Surgical kits'` and `'Vaccination kits'`, respectively. Optimize your model.

If you include the following lines at the end of your code, you should get the output as indicated.

```
Set parameter Username
Set parameter LicenseID to value 2805707
Academic license - for non-commercial use only - expires 2027-04-10
```

```
# Print optimal decision variables Model() object called msf
for var in msf.getVars():
    print(var.VarName, "=", var.X)
```

```
Surgical kits = 13.114754098360653
Vaccination kits = 10.491803278688531
```

b) Next, copy and paste your solution from part a) and rename the Python `Model()` object holding the model from `msf` to `msf_int` in all relevant places. Furthermore, adjust the decision variables so that they can only take integer values. Re-optimize the model.

If you include the following lines at the end of your code, you should get the output as indicated (the solution is actually not unique so other outcomes might also be possible; what is important is that the total number of produced kits should be 23).

```
# Print optimal decision variables from the Model() object called msf_int
for var in msf_int.getVars():
    print(var.VarName, "=", var.X)
```

```
Surgical kits = 12.0
Vaccination kits = 11.0
```

Question 2

A brewery produces beer in Haarlem and Eindhoven (plants), and ships beer to Amsterdam, Amersfoort, Gouda, Den Bosch and Breda (customers).

We can ship beer units from the plants to the customers, taking into account transportation costs, customer demand and plant supply capacities. In the table below we have indicated the following input data:

- Unit transport costs (in euros) from the **plants** (Haarlem, Eindhoven) to the **customers** (Amsterdam, Breda, Gouda, Amersfoort, Den Bosch);
- Demand of each customer: what needs to *at least* be delivered;
- Maximum supply capacity of each plant: what can *at most* be supplied.

	A'dam	Breda	Gouda	A'foort	Den Bosch	Supply
Haarlem	131	405	188	396	485	47
Eindhoven	554	351	479	366	155	63
Demand	28	16	22	31	12	

We define decision variables x_{pc} that model the number of units shipped from plant p to customer c, with the meaning of the indices p and c as follows:

- Plants: $p = 1$ for Haarlem, $p = 2$ for Eindhoven.
- Customers: $c = 1$ for Amsterdam, $c = 2$ for Breda, $c = 3$ for Gouda, $c = 4$ for Amersfoort, $c = 5$ for Den Bosch.

A linear optimization problem that minimizes the total transportation costs subject to the demand and supply constraints is given below.

$$
\begin{array}{lll}
\min & 131x_{11} + 405x_{12} + 188x_{13} + 396x_{14} + 485x_{15} + & \\
& 554x_{21} + 351x_{22} + 479x_{23} + 366x_{24} + 155x_{25} & \\
\text{s.t.} & x_{11} + x_{12} + x_{13} + x_{14} + x_{15} \leq 47 & \textit{supply Haarlem} \\
& x_{21} + x_{22} + x_{23} + x_{24} + x_{25} \leq 63 & \textit{supply Eindhoven} \\
& x_{11} + x_{21} \geq 28 & \textit{demand A'dam} \\
& x_{12} + x_{22} \geq 16 & \textit{demand Breda} \\
& x_{13} + x_{23} \geq 22 & \textit{demand Gouda} \\
& x_{14} + x_{24} \geq 31 & \textit{demand A'foort} \\
& x_{15} + x_{25} \geq 12 & \textit{demand Den Bosch} \\
& x_{11}, x_{12}, x_{13}, x_{14}, x_{15}, x_{21}, x_{22}, x_{23}, x_{24}, x_{25} \geq 0 & \textit{ship nonneg. amounts}
\end{array}
$$

Implement this model in Gurobi with a `Model()` object called `model_beer`. Instantiate all $2 \cdot 5 = 10$ decision variables separately. In the `name` keyword argument of a decision variable, indicate the plant-customer combination it represents; for example, you could define

```
x11 = model_beer.addVar(name='Haarlem-Amsterdam')
```

for the decision variable x_{11} modeling the Haarlem-Amsterdam combination).

If you include the following lines at the end of your code, you should get the output as indicated.

```
for var in model_beer.getVars():
    print(var.VarName, "=", var.X)
```

```
Haarlem-Amsterdam = 28.0
Haarlem-Breda = 0.0
Haarlem-Gouda = 19.0
Haarlem-Amersfoort = 0.0
Haarlem-Den Bosch = 0.0
Eindhoven-Amsterdam = 0.0
Eindhoven-Breda = 16.0
Eindhoven-Gouda = 3.0
Eindhoven-Amersfoort = 31.0
Eindhoven-Den Bosch = 12.0
```

Question 3

In this question we will extend the transportation problem from Question 2 to an arbitrary number of plants and customers. Consider the general transportation problem for m plants and n customers:

$$
\begin{array}{lllll}
\min & \displaystyle\sum_{p=1}^{m}\sum_{c=1}^{n} u_{pc}x_{pc} & & & \textit{total transportation costs} \\
\text{s.t.} & \displaystyle\sum_{c=1}^{n} x_{pc} \leq s_p & \forall p & & \textit{supply available at each p(lant) is } s_p \\
& \displaystyle\sum_{p=1}^{m} x_{pc} \geq d_c & \forall c & & \textit{demand } d_c \textit{ met for each c(ustomer)} \\
& x_{pc} \geq 0 & \forall (p,c) & & \textit{ship nonneg. amounts}
\end{array}
$$

where u_{pc} is the cost of transporting one unit from plant $p \in \{1, \dots, m\}$ to customer $c \in \{1, \dots, n\}$, s_p the supply of plant p and d_c the demand of customer c.

Write a function `transport()` that takes as input a two-dimensional array $u \in \mathbb{R}^{m \times n}$ containing the transportation costs for every plant-customer combination, a one-dimensional array $s \in \mathbb{R}^m_{\geq 0}$ with the supplies of the plants and a one-dimensional array $d \in \mathbb{R}^n_{\geq 0}$ with the demands of the customers. It should output a `Model()` object that implements the above problem; do not yet optimize the model in this function!
Hint: Section 5.2.3 on the use of double-indexed variables is useful.

Test your function on the input from Question 2 by executing the following code below your function.

```
u = np.array([[131, 405, 188, 396, 485],[554, 351, 479, 366, 155]])
s = np.array([47, 63])
d = np.array([28, 16, 22, 31,12])

# Define Model() object through transport()
question2_data = transport(u,s,d)

# Optimize the model
question2_data.optimize()

# Print optimal decision variables
for var in question2_data.getVars():
    print(f'{var.VarName} = {var.X}')
```

```
combinations[0,0] = 28.0
combinations[0,1] = 0.0
combinations[0,2] = 19.0
combinations[0,3] = 0.0
combinations[0,4] = 0.0
combinations[1,0] = 0.0
combinations[1,1] = 16.0
combinations[1,2] = 3.0
combinations[1,3] = 31.0
combinations[1,4] = 12.0
```

Question 4

Consider a knapsack problem in which we have $k = n^2 \in \mathbb{N}$ items $I = \{1, \dots, k\}$ that all have a weight w_i and a value v_i for $i \in I$. We want to select a subset of items whose total weight is at most W, whilst their total value is maximized at the same time. Furthermore, we can select at most r items from the first group of n items $\{1, \dots, n\}$, at most r from the second group of n items $\{n+1, \dots, 2n\}$ etc.

This problem can be modelled with the following binary linear optimization problem, where $x_i = 1$ if item i is included, and $x_i = 0$ otherwise.

$$
\begin{aligned}
\max \quad & \sum_{i=1}^{n^2} v_i x_i & & & \\
\text{s.t.} \quad & \sum_{i=a\cdot n+1}^{(a+1)\cdot n} x_i & \leq & r & \text{for } a = 0, \dots, n-1 \\
& \sum_{i=1}^{n^2} w_i x_i & \leq & W & \\
& x_i & \in & \{0,1\} & \text{for all } i = 1, \dots, n^2
\end{aligned}
$$

a) Define a function `knapsack(n,w,v,W,r)` that takes as input $n, r \in \mathbb{N}, v, w \in \mathbb{R}^{n^2}_{\geq 0}$ and $W \in \mathbb{R}_{\geq 0}$, and returns the implemented problem as a `Model()` object. Do not optimize the model inside this function!

Test your function by copying and pasting the code below (the solution is not unique so other outputs might also be possible; check they have the same objective function as the displayed optimal solution here).

```
# 16 items divided into four consecutive groups of four items
n = 4
w = np.array([2, 5, 4, 7,
              3, 6, 2, 8,
              5, 4, 7, 3,
              6, 2, 5, 4])

v = np.array([6, 12, 9, 15,
              8, 14, 5, 18,
              11, 10, 16, 7,
              13, 6, 12, 9])

W = 25  # Maximum total weight
r = 2   # At most two items from each group

model_knapsack = knapsack(n, w, v, W, r)
model_knapsack.optimize()

for var in model_knapsack.getVars():
    print(var.VarName, "=", var.X)
```

```
items[0] = 1.0
items[1] = 1.0
items[2] = 0.0
items[3] = -0.0
items[4] = 1.0
items[5] = 1.0
items[6] = 0.0
items[7] = 0.0
items[8] = -0.0
items[9] = 1.0
```

```
items[10] = 0.0
items[11] = 1.0
items[12] = -0.0
items[13] = 1.0
items[14] = 0.0
items[15] = 0.0
```

b) Is it necessary to have n as an input argument of `knapsack(n,w,v,W,r)`, or can you implement the same problem without n as input, i.e., `knapsack(w,v,W,r)`? You may still assume that $k = n^2$ to answer this question.

Question 5

A university wants to assign bachelor students to thesis topics. Each teacher proposes exactly one topic, and each topic is proposed by exactly one teacher. Assigning a student to a topic therefore also determines their supervisor.

There are m students and n teachers. Every student has given every teacher's topic a grade between 1 and 10, where a higher grade indicates a stronger preference. These grades are stored in a matrix $G \in [1, 10]^{m \times n}$, where G_{ij} is the grade given by student i to the topic proposed by teacher j.

The university wants to maximize the total preference score, subject to the following constraints:

- Every student must be assigned to exactly one teacher.
- Each teacher can supervise at most $U = 5$ students.
- For any two teachers, the numbers of students they supervise may differ by at most 2.

We use student indices $i = 0, \dots, m-1$ and teacher indices $j = 0, \dots, n-1$. Define binary decision variables

$$x_{ij} = \begin{cases} 1 & \text{if student } i \text{ is assigned to teacher } j, \\ 0 & \text{otherwise.} \end{cases}$$

The problem can be formulated as the following binary linear optimization problem:

$$\begin{aligned}
\max \quad & \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} G_{ij} x_{ij} && \text{total preference score} \\
\text{s.t.} \quad & \sum_{j=0}^{n-1} x_{ij} = 1 && \forall i = 0, \dots, m-1, \\
& \sum_{i=0}^{m-1} x_{ij} \leq U && \forall j = 0, \dots, n-1, \\
& \sum_{i=0}^{m-1} x_{ip} - \sum_{i=0}^{m-1} x_{iq} \leq 2 && \forall\, 0 \leq p < q < n, \\
& \sum_{i=0}^{m-1} x_{iq} - \sum_{i=0}^{m-1} x_{ip} \leq 2 && \forall\, 0 \leq p < q < n, \\
& x_{ij} \in \{0, 1\} && \forall i, j.
\end{aligned}$$

The two constraints for each pair of teachers ensure that their supervision loads differ by at most 2 in either direction. Teachers who supervise no students are included in this comparison.

a) Write a function `thesis_assignment()` that takes as input a two-dimensional array `G` containing the preference grades and the maximum number of students per teacher, `U`. Return a `Model()` object implementing the problem. Do not optimize the model inside the function.
Hint: You can store each teacher's supervision load as a `quicksum()` expression and reuse these expressions in the balancing constraints.

Solution

b) Test your function on the following data for 12 students and 4 teachers.

```
# Rows: students; columns: teachers' topics
G = np.array([
    [10, 8,  3,  2],
    [10, 7,  4,  3],
    [ 9, 8,  3,  4],
    [10, 9,  2,  3],
    [ 9, 7,  4,  2],
    [10, 8,  3,  4],
    [ 9, 8,  4,  3],
    [ 4, 3, 10,  6],
    [ 3, 4,  9,  5],
    [ 5, 3, 10,  7],
    [ 3, 4,  6, 10],
    [ 4, 3,  5,  9]
])

U = 5

model_thesis = thesis_assignment(G, U)
model_thesis.optimize()

# Print the selected student-teacher combinations
for var in model_thesis.getVars():
    if var.X > 0.5: # Print only variables with value 1
        print(var.VarName, '=', round(var.X))
```

```
assignment[0,0] = 1
assignment[1,0] = 1
assignment[2,1] = 1
assignment[3,1] = 1
assignment[4,0] = 1
assignment[5,0] = 1
assignment[6,1] = 1
assignment[7,2] = 1
assignment[8,2] = 1
assignment[9,2] = 1
assignment[10,3] = 1
assignment[11,3] = 1
```